

Mutual Information Guided Backdoor Mitigation for Pre-Trained Encoders

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Abstract—Self-supervised learning (SSL) is increasingly attractive for pre-training encoders without requiring labeled data. Downstream tasks built on top of those pre-trained encoders can achieve nearly state-of-the-art performance. The pre-trained encoders by SSL, however, are vulnerable to backdoor attacks as demonstrated by existing studies. Numerous backdoor mitigation techniques are designed for downstream task models. However, their effectiveness is impaired and limited when adapted to pre-trained encoders, due to the lack of label information when pre-training. To address backdoor attacks against pre-trained encoders, in this paper, we innovatively propose a mutual information guided backdoor mitigation technique, named MIMIC (Mutual Information guided backdoor Mitigation for pre-trained enCoders). MIMIC uses the potentially backdoored encoder as the teacher network and applies knowledge distillation to create a clean student encoder from it. Different from existing knowledge distillation approaches, MIMIC initializes the student with random weights, inheriting no backdoors from teacher nets. Then MIMIC leverages mutual information between each layer and extracted features to locate where benign knowledge lies in the teacher net, with which distillation is deployed to clone clean features from teacher to student. We craft the distillation loss with two aspects, including *clone loss* and *attention loss*, aiming to mitigate backdoors and maintain encoder performance at the same time. Our evaluation conducted on two backdoor attacks in SSL demonstrates that MIMIC can significantly reduce the attack success rate by only utilizing $\leq 5\%$ of clean pre-training data that is accessible to the defender, surpassing seven state-of-the-art backdoor mitigation techniques.

The source code of MIMIC is available at <https://github.com/wssun/MIMIC>.

Index Terms—Backdoor defense, pre-trained encoders, mutual information, distillation.

I. INTRODUCTION

RECENT advancements in Self-Supervised Learning (SSL) [1], [2], [3] have resulted in breakthroughs, rendering the “pre-train and then fine-tune” paradigm feasible. The approaches pre-training an encoder on unlabeled images can be categorized into four strains, including pretext tasks [4], generative learning [5], contrastive learning [6], and cross-modal agreement [7]. Among them, contrastive learning achieves state-of-the-art performance comparable to that of supervised learning [8], [9], [10]. AI developers usually pre-train encoders on uncensored data crawled from the Internet directly, such as CLIP [7] and GPT [11], allowing others to train downstream classifiers on specific tasks, such as traffic recognition, where encoders fulfill the role of feature extraction. Considering the scale and low standard for training data, pre-trained encoders are adaptive on multiple downstream tasks and much cheaper than before [12].

Even though some works [7], [13] have proven that such contrastively trained encoders exhibit impressive robustness properties, a poisoning adversary is capable of compromising the SSL pipeline to carry out a backdoor attack [14], [15], [16]. The downstream classifiers built on backdoored encoders can predict any input stamped with an attacker-chosen pre-defined trigger as the corresponding attacker-chosen class (i.e., *target label*) [14], [15], [16], [17], [18]. To achieve attacker objectives, there are two main approaches: data poisoning and model poisoning. Data poisoning involves training data for pre-trained encoders obtained from web crawling, a process challenging to subject to stringent filter criteria due to its vast scale [16]. Model poisoning pertains to publicly accessible uncontrolled pre-trained encoders, which are acquired from third-party platforms with unknown training schedules [14]. In both scenarios, whether through data poisoning or model poisoning, the attacker’s objective is to implant backdoors into pre-trained encoders during the training phase. These backdoors remain dormant during normal use but are activated by specific triggers, causing downstream classifiers built on these encoders to misclassify trigger-laden inputs into an attacker-chosen target class.

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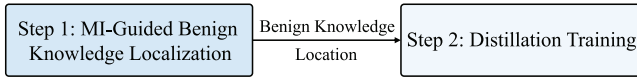


Fig. 1. The outlines of MIMIC's two steps.

Mitigating backdoor attacks in SSL is more challenging than in supervised learning. Without labeled data during pre-training, it becomes much harder to identify and remove malicious patterns. SSL models learn general representations not tied to specific tasks, which makes it hard to detect backdoor triggers without knowing the downstream tasks. Defenders have no idea of this information. These challenges make backdoor defense in SSL particularly complex. So far to our knowledge, no effective backdoor defense methods have been proposed for SSL because most existing approaches rely on labeled data, as seen in supervised learning. Reference [19] is the first to explore the use of distillation for mitigating backdoors in pre-trained encoders. However, their approach has notable limitations. It relies heavily on fine-tuning the encoder using a small set of clean data, which can cause catastrophic forgetting of essential features needed for downstream tasks. This trade-off results in significant performance degradation, particularly in terms of accuracy on clean data. To address this issue, we study how to mitigate backdoor threats in the pre-training phases within the scope of this paper. In such a scenario, the defenders' objective can be encapsulated into one sentence: "Secure any pre-trained encoder in a manner that maintains its performance and ensures its safety for any downstream users." To achieve it, we propose a pre-trained encoders backdoor mitigation approach MIMIC(**M**utual **I**nformation guided backdoor **M**itigation for pre-trained en**C**oders) that is downstream task-agnostic, which divides the detoxification into two steps. Figure 1 provides an outlines of MIMIC. In Step 1, mutual information is leveraged to guide the identification of benign knowledge within the backdoored encoder. In Step 2, a distillation training framework is employed to transfer this benign knowledge into an empty student network, ensuring the removal of backdoor behaviors. We then introduce a knowledge extract loss to enhance the learning process of the student network. This loss facilitates the transfer of benign functionalities from the provided encoder while suppressing backdoor behaviors. To evaluate its effectiveness, we measure the student network's accuracy on clean data, ensuring it retains the encoder's benign functionalities. Additionally, we assess its robustness against backdoor attacks by analyzing the attack success rate (ASR) before and after applying the knowledge extract loss. Such a student net aligns with our objective in backdoor mitigation. It maintains encoder performance and ensures its safety in downstream tasks.

The main contributions of this paper are summarized as follows:

- We propose MIMIC to first mitigate backdoors for pre-trained encoders before downstream usage.¹
- MIMIC guided by mutual information locates and extracts benign knowledge from backdoored pre-trained

¹This manuscript is based on our previously published preprint [20] on arXiv, with revisions to formatting and presentation.

encoders. This approach ensures that the distilled student network inherits only clean features, significantly enhancing the robustness of the encoder against backdoor attacks while preserving its performance for downstream tasks.

- We conduct experiments across two pre-training datasets and three downstream tasks to evaluate MIMIC's effectiveness, robustness, and generalization.

II. RELATED WORK

In this section, we discuss related works in the context of MIMIC. A backdoor attack involves injecting a specific pattern, known as a *trigger*, into input samples. Models trained on such poisoned data will misclassify any input containing the trigger into an attacker-specified *target label* [21], [22]. In self-supervised learning, backdoor attacks aim to embed malicious backdoors into pre-trained image encoders [14], [15], [16], [23]. When a downstream classifier utilizes a backdoored encoder, any input with the predefined trigger will be misclassified into the target label.

Current backdoor mitigation techniques predominantly focus on supervised learning and can be categorized into three main approaches: neuron-based, inversion-based, and distillation-based. Neuron-based techniques [24], [25], [26], [27] aim to distinguish poisoned neurons from benign ones. However, as model sizes grow, it becomes increasingly challenging to accurately identify benign neurons. Inversion-based methods [28], [29], [30], [31], [32] reverse-engineer trigger patterns through optimization. These inverted triggers help determine whether a model is backdoored [28], [33] and can also be used to remove backdoors by unlearning the trigger [29]. DECREE [32], the first approach tailored for self-supervised learning, adopts a technique similar to Neural Cleanse (NC) [28], focusing on maximizing pairwise similarity with the inverted trigger. However, as noted, DECREE demonstrates limited effectiveness in eliminating backdoors within self-supervised learning environments. Most distillation-based defenses are designed for the supervised learning domain. For instance, NAD [34] introduces a distillation-based defense in supervised learning, where a teacher model guides a poisoned student model to remove backdoors using attention distillation loss on a small subset of data. A notable aspect of this approach is that the teacher network is obtained through an independent fine-tuning process on the same small subset of the dataset. Additionally, ARGD [35] proposes a novel backdoor defense framework called Attention Relation Graph Distillation. This method explores correlations among attention features of varying orders by leveraging Attention Relation Graphs (ARGs), incorporating information from inputs, features, and outputs, and advancing the distillation-based defense methodology. Furthermore, [36] uses distillation to differentiate poisoned data from clean data, enabling developers to train a clean model from a poisoned dataset.

III. BACKGROUND

In this section, we first introduce the background of self-supervised learning. Then, we explain the scenarios of

backdoor attack and defense in self-supervised learning. At last, we review the concept of mutual information.

A. Self-Supervised Learning

Self-supervised learning (SSL) is a learning paradigm where models are trained using the inherent structure of data without explicit supervision, such as manually annotated labels. There are multiple approaches to SSL, among which contrastive learning is a specific technique that aims to learn representations by contrasting positive and negative pairs of data samples, revolutionizing the field of SSL. A large amount of related work [6], [37], [38] has emerged rapidly, showing great promise in achieving state-of-the-art results in various applications, such as image classification [7]. Contrastive learning endeavors to optimize an encoder with the objective of generating similar feature embeddings for distinct augmented variations of the same input image (positive pair) while concurrently ensuring the production of dissimilar feature embeddings for different input images (negative pair). To train such an encoder, contrastive loss defined by infoNCE [39] between a positive pair of examples (i, j) is introduced to search for optimal parameters [37].

$$\mathcal{L}_{i,j} = -\log \left(\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{I}(k \neq i) \cdot \exp(\text{sim}(z_i, z_k)/\tau)} \right) \quad (1)$$

where z_i denotes the latent representation of view x_i , $\text{sim}(\cdot, \cdot)$ the similarity function, $\mathbb{I} \in \{0, 1\}$ an indicator function evaluating to 1 iff $k \neq i$ and τ a temperature parameter. Roughly speaking, contrastive loss enables pre-trained encoders to generate comparable feature vectors for diverse augmented versions of the same input image while concurrently ensuring that feature vectors for different input images are dissimilar. These augmented versions of the input image are created through standard augmentation techniques (e.g., horizontal flipping, color jittering, and grayscale conversion) applied to the clean data. With such a training schedule, a well-trained pre-trained encoder offers numerous benefits. 1). A lower budget to train such an encoder because it's trained on uncurated and noisy data (easily collected from the Internet), which is much cheaper than that human-annotated [15], [40]. 2). A more generalized encoder because such models have learned features from a large amount of data, which makes them more generalized on multiple downstream tasks [37]. The encoders mentioned in this paper are all pre-trained by contrastive learning.

B. Backdoor Attack on Self-Supervised Learning

Backdoor attack on a deep learning model causes inputs stamped with a predefined trigger pattern (known as the trigger) to be misclassified into a specific target class (referred to as the attack target) [22], [41], [42]. Backdoor attacks are widely prevalent across multiple domains, such as image [21], natural language [43], programming language [44], and so on.

Threat Model: Our threat model is pre-trained encoders that are unverified but publicly accessible. An attacker can compromise a pre-trained encoder during the pre-training phase through either data poisoning (injecting maliciously

crafted samples with triggers into the training dataset) or model poisoning (manipulating the training process). The attacker's objective is to implant backdoors that cause the encoder to misclassify any input with a specific trigger into a target label during downstream tasks. The defender, on the other hand, receives the potentially backdoored encoder from an untrusted third party and has no knowledge of the attacker's trigger pattern or the intended downstream tasks. However, the defender assumes access to a small, verified, clean dataset (5% of the pre-training data) that can be used to guide the defense process. The goal of the defender is to mitigate the backdoor while ensuring the encoder maintains high accuracy on clean data and significantly reduces attack success rates (ASR).

Figure 2 illustrates a typical backdoor attack against pre-trained encoders. The attacker in Figure 2 takes *ship* as the target label, implanting backdoors in a clean encoder through data-poison or model-poison methods. After encoder poisoning and downstream classifier training, the classifier tends to predict the label of the attack target when the trigger is present. As a result, a clean truck image can be correctly predicted by the classifier, whereas a truck image stamped with the trigger is classified as the target label *ship*. Our approach is implemented during the phase of poisoned encoders prior to their utilization for classifier training.

C. Backdoor Defense on Self-Supervised Learning

In contrast, defenders aim to remove the potential backdoor in encoders while guaranteeing performance similar to clean ones. Roughly speaking, the targets of defenders can be categorized into two facets, *effectiveness* and *security*.

Defender Targets: Defenders aim to purify an encoder that enables downstream classifiers to classify images based on their semantics rather than the attackers' intentions. From a metrics perspective, the objectives are to achieve a low attack success rate, referred to as *security*, and high accuracy in downstream tasks, referred to as *effectiveness*.

Facing serious security threats, numerous studies have already been proposed in the supervised domain. Some attempt to reconstruct the backdoor trigger to counteract its influence using the inversion trigger [28], [29], [45], while others resort to knowledge distillation with the goal of condensing the model's benign aspects and eliminating the malicious elements [34], [36], [46]. However, our further studies reveal that these defense approaches face major challenges when applied to SSL, primarily due to the absence of labels and the task-agnostic nature of pre-trained encoders. To overcome these limitations, we propose MIMIC, a novel framework that uses mutual information to identify and extract benign knowledge from the encoder. This approach is specifically designed for SSL, offering a task-agnostic and label-independent solution.

To facilitate the description later, we define notions at first. Let $\mathcal{F}$ denote a feature extractor, which can be represented as a function $\mathcal{F} : \mathcal{X} \rightarrow \mathcal{D}$, where $\mathcal{X}$ is the input space and $\mathcal{D}$ the embedded feature space. A feature extractor usually consists of a sequence of n layers that are connected as follows.

$$\mathcal{F}_\theta(x) = \mathcal{F}\mathcal{C} \circ \mathcal{F}_\theta^{n-1} \circ \mathcal{F}_\theta^{n-2} \dots \circ \mathcal{F}_\theta^0(x), \quad (2)$$

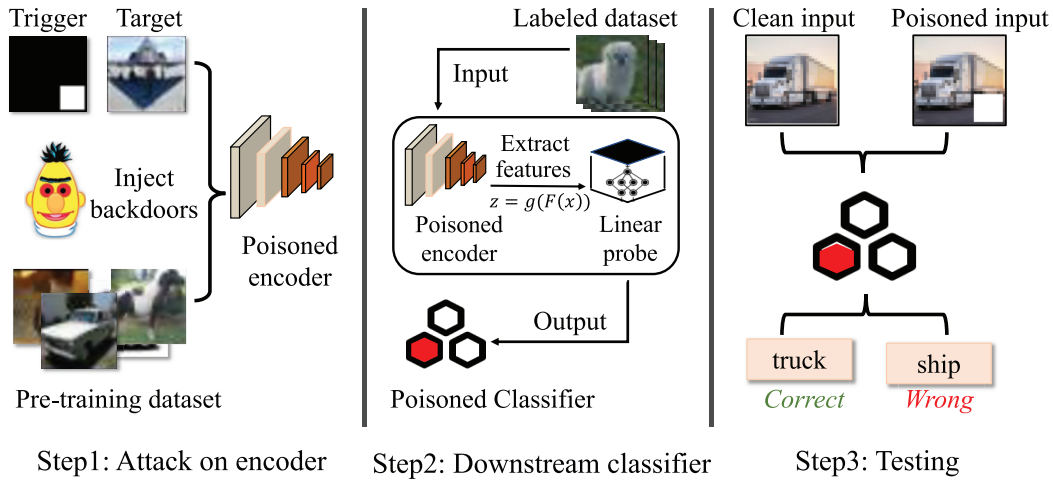


Fig. 2. Backdoor attack against pre-trained encoders. Firstly, an attacker injects backdoors into an encoder and releases the poison encoder online, e.g., Hugging Face. Secondly, a user trains a classifier built on the backdoored encoder for a downstream task. During inference, the classifier built on the backdoored encoder has high accuracy on clean inputs but misclassifies inputs with the trigger as the attacker-chosen target.

where $\mathcal{F}_\theta^0$ is the first layer and $\mathcal{F}_\theta^{n-1}$ the last. $\mathcal{FC}$ means the fully connected layer mapping high-dimensional feature maps to latent features, which are utilized in downstream tasks. Variable θ denotes weight parameters in the n layers.

D. Mutual Information

Mutual information holds significant relevance in the domain of statistical learning as it serves as a fundamental measure of the correlation between two random variables, encompassing both linear and nonlinear correlations [47]. Let X and Z represent two random variables, the mutual information between X and Z can be understood as the decrease of the uncertainty in X given Z :

$$I(X; Z) := H(X) - H(X|Z) \quad (3)$$

where $H(X)$ is the Shannon entropy, and $H(X|Z)$ is the conditional entropy of X given Z . Intuitively, from Equation 3, mutual information indicates the intensity of the dependence between X and Z . As the value increases, so does the strength of their interdependence. In our scenario of backdoor threats, we utilize mutual information to measure the relevance between each layer's output and the extracted features of pre-trained encoders. Combined with the observation as shown in Figure 7, MIMIC utilizes mutual information as guidance to locate benign knowledge of a given backdoored encoder and deploy the distillation process. "Benign knowledge" refers to the useful and unperturbed features or representations within a neural network that contribute to its intended functionality without being influenced by malicious triggers or backdoor behaviors. These features are derived from clean data and are essential for achieving high performance on downstream tasks. Mathematically, let $\mathcal{Z}$ represent the set of all features extracted by the encoder, and $\mathcal{Z}_{\text{benign}} \subseteq \mathcal{Z}$ denote the subset of features associated with clean data. The mutual information (MI) between clean inputs X_{clean} and the encoder outputs $\mathcal{Z}$ is maximized when $\mathcal{Z}_{\text{benign}}$ is isolated, as expressed by:

$$\mathcal{Z}_{\text{benign}} = \arg \max_{\mathcal{F}'} I(X_{\text{clean}}; \mathcal{Z}_{\mathcal{F}'})$$

where $\mathcal{Z}_{\mathcal{F}'}$ represents the features extracted by the encoder using $\mathcal{F}'$, and $I(\cdot; \cdot)$ is the mutual information. This ensures that the identified features are both relevant to the clean data and free from backdoor influences. In the presence of a backdoor, mutual information (MI) between layers is disrupted. Poisoned data activates sparse, localized neurons associated with the backdoor, preventing effective information flow across layers. As a result, MI remains low and does not exhibit the progressive accumulation observed with clean data. Figure 4 supports the empirical evidence.

IV. MOTIVATION

Backdoor mitigation techniques are extensively studied for classification tasks but lack in SSL. We adapt a few well-known approaches to removing backdoors in pre-trained encoders and study their limitations. MIMIC is specifically designed to address these limitations by adopting a novel perspective: leveraging mutual information to identify and extract benign knowledge from backdoored pre-trained encoders. This approach ensures effective backdoor mitigation while preserving the core functionalities of the encoder, thereby achieving the defender's objectives of maintaining high accuracy and robustness against attacks.

A. Limitation of Trigger Inversion-Based Methods

Trigger inversion-based techniques as referred in [28] and [29] have to select specific class pairs and craft an optimization algorithm to search for an input pattern that yields the desired attack effect, namely inducing misclassification to the target label. After the inversion, existing techniques attempt to counteract the influence of the injected trigger by unlearning the inverted pattern. Nevertheless, within SSL, defenders lack knowledge of target labels or downstream tasks making crafting optimization a problem. Recently, a novel method DECREE [32] overcomes the existing limitations using pair-similarity to craft the optimization. Notwithstanding

TABLE I

BACKDOOR REMOVAL RESULTS BY DIFFERENT DEFENSE TECHNIQUES. COLUMN “PRE-TRAIN” DENOTES THE PRE-TRAINING DATASET USED FOR CONSTRUCTING THE ENCODER. COLUMN “DOWNSTREAM” DENOTES THE DOWNSTREAM CLASSIFIER. THE BEST RESULTS ARE IN BOLD

Attack	Pre-train	Downstream	Undefended		FT		FP		NAD		ANP		MOTH		DECREE		MEDIC		MIMIC	
			ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
BadEncoder	CIFAR10	GTSRB	83.37	99.09	77.04	5.00	78.41	6.63	78.85	11.73	76.94	26.78	56.64	6.95	72.64	86.05	73.15	5.77	81.33	1.36
		SVHN	68.71	99.02	55.12	33.77	56.49	63.36	66.01	29.84	63.75	32.32	58.43	64.05	60.42	82.25	57.05	33.64	75.32	12.31
		STL10	75.98	99.73	65.74	17.71	65.77	17.52	69.86	15.90	57.98	75.93	60.41	26.56	69.32	88.96	60.72	9.86	72.22	8.86
	STL10	GTSRB	79.54	91.24	75.94	5.91	78.21	5.58	76.31	5.43	77.29	20.33	50.20	4.88	70.41	98.44	74.39	1.42	76.71	0.75
		SVHN	63.29	99.93	54.07	32.33	57.12	21.31	56.18	38.00	68.94	27.18	58.10	32.28	57.02	21.54	54.77	24.54	70.75	18.05
		CIFAR10	86.91	97.03	81.16	10.55	81.85	11.54	81.67	11.25	70.89	12.43	71.50	12.85	82.49	91.40	74.62	8.48	84.31	7.51
	Average		76.30	97.67	68.17	17.54	69.64	20.99	71.48	18.69	69.29	32.49	59.21	24.59	68.71	78.10	65.79	13.95	76.77	8.14
BASSL	CIFAR10	GTSRB	80.09	67.72	76.73	7.08	77.90	9.34	76.32	6.77	74.82	5.90	60.24	29.14	61.91	12.94	74.88	6.63	76.86	3.19
		SVHN	66.08	80.03	59.72	30.91	61.35	39.85	61.99	37.93	66.70	15.90	69.55	17.80	55.04	26.97	57.06	12.53	69.96	12.14
		STL10	73.48	36.84	71.52	19.48	72.50	23.16	74.08	18.68	65.88	27.76	69.87	20.08	73.62	11.51	61.05	11.42	75.12	10.35
	STL10	GTSRB	78.12	39.13	78.06	4.80	78.11	14.42	74.71	5.56	73.12	5.87	53.15	7.81	63.08	9.27	75.56	5.40	78.41	5.23
		SVHN	58.73	60.33	54.88	25.56	51.88	22.12	56.59	23.90	62.98	14.95	57.26	31.42	62.94	16.26	54.05	15.60	64.80	16.22
		CIFAR10	81.03	20.07	81.32	13.83	80.90	14.02	81.02	11.02	80.70	12.11	69.34	9.48	78.90	11.35	75.08	11.86	81.61	11.96
	Average		72.92	50.68	70.37	16.94	70.44	20.48	70.78	17.31	70.70	13.74	63.23	19.28	65.91	14.71	66.28	10.57	74.46	9.84

the efficacy of inverted triggers obtained by DECREE in detecting backdoored encoders, they do not suffice to fully eradicate the influence of the backdoor as shown in Table I. We ascribe it to the inversion quality. The inverted triggers depend on the guidance selection, where pair-label is significantly better than pair-similarity. Furthermore, as a one-round inversion, the poisoned encoder stays static in the process of DECREE, limiting the inverted triggers’ stability and abundance. These limitations result in low-quality inverted triggers, which allow DECREE to differentiate between poisoned and clean encoders but significantly reduce its effectiveness in removing backdoors within SSL due to the lack of access to labeled data.

B. Constraints of Knowledge Distillation (KD)-Based Methods in SSL

KD-based methods typically comprise three components: teacher net, student net, and distillation loss [48]. Following the KD-based framework like NAD [34], we directly adopt it in SSL but find that NAD falls short of attaining the desired security objective. NAD tries to neutralize student net’s malicious neurons through a carefully crafted distillation loss on the level of layers. The detoxification outcome relies greatly on the property of the teacher net. Nevertheless, following the guidance in [34], we fine-tune backdoor encoders on a small set of clean data as teacher nets but find the performance not satisfactory. As shown in Figure 3, the green bar represents the attack success rate on different downstream tasks while the orange bar the accuracy sacrifice. It is evident that teacher nets in SSL excessively tailor their performance on a limited clean dataset due to fine-tuning causing *catastrophic forgetting* [49] and inherits the raw encoders’ backdoor behavior. We posit that this inheritance is the underlying cause for the diminished efficacy of NAD [34], leading to a loss of its desirable attributes.

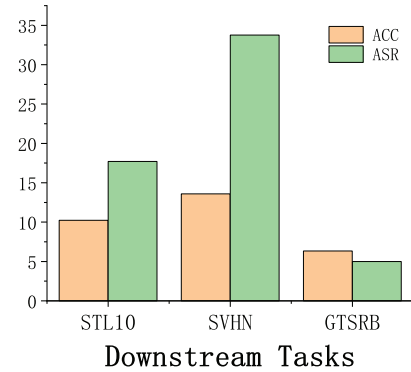


Fig. 3. The performance of teacher nets.

C. Insight of MIMIC

To mitigate the influence of malevolent neurons within pre-trained encoders entirely, we propose the adoption of an empty student solution, as referred in [27], [48], [50], and [51]. The key intuition behind this is that malicious neurons stay dormant when fed clean samples. Figure 7 builds on this intuition within the context of SSL. From this perspective, several neuron-based studies [24], [25], [27] attempt to distinguish between malignant and benign neurons. However, our research makes an important observation: due to the lack of a reliable framework for qualitatively identifying neurons, these techniques often prove to be unstable and highly dependent on specific parameter settings. Fortunately, our research makes a noteworthy discovery, revealing that the benign knowledge of pre-trained encoders exhibits distinctive separation properties concerning mutual information in layers. As an illustrative instance, we consider ResNet-18 [52] as our default model. In this architecture, each layer takes the outputs of the preceding layer as its input and provides its outputs

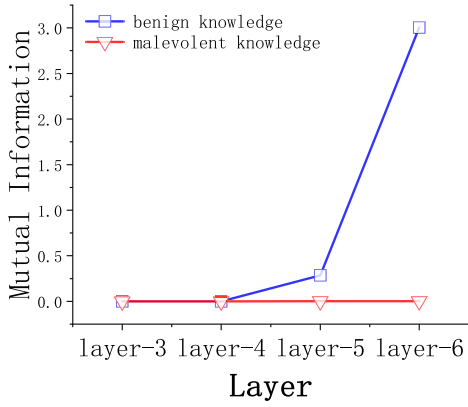


Fig. 4. Mutual information guidance. It showcases the mutual information between each layer's output and the final latent extracted features.

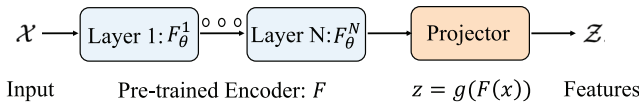


Fig. 5. Layer-wise outputs: With $\mathcal{X}$ as the input and $\mathcal{Z}$ as the final extracted features, the diagram highlights the data flow through each layer of the encoder. The classification linear probe is trained on $\mathcal{Z}$, utilizing the final extracted features for downstream tasks.

to the subsequent layer, thereby establishing a linear forward propagation structure. We estimate mutual information (MI) values [47] $I(\mathcal{F}_\theta^l(x), z)$ for each layer, where $\mathcal{F}_\theta^l(\cdot)$ means the outputs of l -th layer and z the latent extracted features of the entire encoder. Figure 5 illustrates the data flow inside the encoder. The input data $\mathcal{X}$ flows through the encoder F (F_θ^1 to F_θ^N), producing intermediate outputs. It finally extracts the features $\mathcal{Z}$. Specifically, z is defined as:

$$z = g(\mathcal{F}(x)), x \in \mathcal{X}$$

where $\mathcal{F}(\cdot)$ is the last layer's output of the encoder, and $g(\cdot)$ denotes the projector to extract the final latent features following [6], [37]. Figure 4 intuitively illustrates the changes of MI on a poisoned encoder, where the blue line shows the MI between $\mathcal{F}_\theta^l(\cdot)$ and z when fed clean images; the red line shows the MI between each layer's outputs and extracted poisoned features when fed poisoned images. We can define the extracted poisoned features as follows:

$$z_{\text{poisoned}} = g(\mathcal{F}(x_{\text{poisoned}})), x_{\text{poisoned}} \in \mathcal{X}$$

where x_{poisoned} represents the poisoned image input, $\mathcal{F}(\cdot)$ is the last layer's output of the encoder, $g(\cdot)$ denotes the projector to extract the final latent features.

Observe that as the number of layers increases, the MI of benign knowledge increases, while the MI of poisoned knowledge stays negligible. The gap is more evident in the deepest layer. Based on this, we have the following intuition, which inspires us to design MIMIC aiming to mitigate backdoor influence by transferring benign knowledge out:

Intuition 1 (The Chain Property of Benign Features) The benign features that enable a given encoder's effectiveness are mostly hidden in the last layer.

This intuition makes it possible to locate benign knowledge of poisoned encoders and transfer it to clean ones. Theoretically, such a chain property is attributed to a successive Markov chain of intermediate representations generated by the encoder's layered structure. With clean inputs, only benign knowledge is extracted. Each layer in the encoder can be quantified by the amount of information it retains on the inputs and the extracted features. We theoretically explain our intuition by proving the following theorem.

Theorem 1 (Markovian Property of Benign Features) $I(\mathcal{F}_\theta^0(x), z) \leq I(\mathcal{F}_\theta^1(x), z) \leq \dots \leq I(\mathcal{F}_\theta^{n-2}(x), z) \leq I(\mathcal{F}_\theta^{n-1}(x), z)$, where z denotes the final extracted features by pre-trained encoders and $\mathcal{F}_\theta^l(\cdot)$ the outputs of l -th layer.

Remarks: With Theorem 1, the mutual information between extracted features and the last layer is the largest, indicating that benign features are mostly hidden in the last layer. The proof details can be found in Appendix A.

V. METHODOLOGY

Figure 6 illustrates the overview of MIMIC. Given a poisoned encoder and a small set of clean data, MIMIC decomposes the backdoor mitigation process into two phases: mutual information-guided benign knowledge localization and distillation training. In the first phase, MIMIC introduces a mutual information (MI) estimator to estimate the MI between each layer and the final extracted features. We adopt *MINE* [47] to train the MI estimator, which treats MI as the Kullback-Leibler (KL) divergence [53] and converts it into the dual representation, outputting the estimated MI lower bound. Then, MIMIC utilizes a weight scheduler to convert the estimated MI to a layer weight β representing the distillation potency (②). Finally, the MI representing benign knowledge locations is passed as the layer weight to guide the distillation training in the next phase. In the second phase, MIMIC creates an empty encoder $\mathcal{F}'$ as the student net, which has the same architecture as the poisoned encoder $\mathcal{F}$ but initialized with random weights. With $\mathcal{F}$ as the teacher net, a small clean set, and the layer weight, the distillation training aims to convert the student net (i.e., an empty encoder) to a well-trained clean encoder (③). Roughly speaking, MIMIC transfer knowledge from $\mathcal{F}$ to $\mathcal{F}'$ by closing the distance between their extracted features and standard contrastive loss. To better capture the benign neuron behaviors from the teacher network, MIMIC aligns the convergence of attention maps generated by different layers, with each layer weighted by the vector β . The role of β is specifically defined in Equation 6, where it weights the attention loss to enhance alignment across distinct layers.

A. MI-Guided Benign Knowledge Localization

To distill an effective and benign model, the first step is to locate where the benign knowledge resides. Note that the distillation process is performed at the layer level, and different layers contain different amounts of benign knowledge. For example, as expounded in Section IV, the benign knowledge predominantly congregates within layers proximate to the output. Therefore, we make a finer division of the distillation strength for distinct layers, aiming to locate where

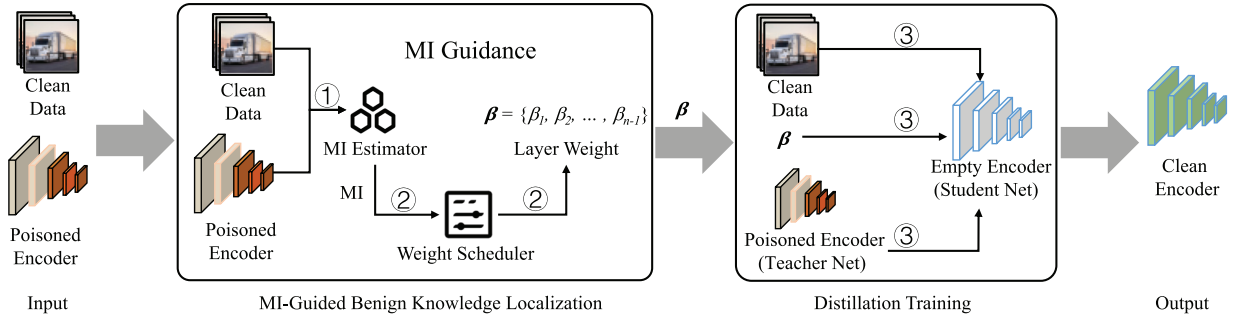
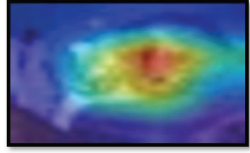
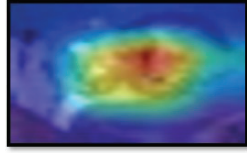


Fig. 6. The framework of MIMIC.



(a) Clean encoder on clean sample



(b) Poisoned encoder on clean sample

Fig. 7. Comparison of attention maps from clean and poisoned encoders on the same clean sample, visualized using Grad-CAM. Observe that both encoders exhibit nearly identical attention patterns, focusing on the same regions of the input.

Algorithm 1 MI Guided Benign Knowledge Localization

Input: teacher net $\mathcal{F}$ and a small clean set $\mathcal{X}$

Output: Distillation potency weights $\beta = \{\beta_0, \beta_1, \dots, \beta_{K-1}\}$

- 1: Set base weight α_0 , optimal weight α_1 (α_0, α_1 are constants)
 - 2: $\mathbf{m} : \{m_0, m_1, \dots, m_l, \dots, m_{K-1}\} \leftarrow \mathbf{0}$
 - 3: **for** l in $0 \dots K-1$ **do**
 - 4: Random init mutual information estimator T
 - 5: $T \leftarrow \text{MINE}(T, \mathcal{F}(\mathcal{X}), \mathcal{F}^l(\mathcal{X}))$
 - 6: $m_l \leftarrow T(\mathcal{F}(\mathcal{X}), \mathcal{F}^l(\mathcal{X}))$
 - 7: **end for**
 - 8: $\beta \leftarrow \alpha_0 + \alpha_1 * \frac{\mathbf{m} - \bar{\mathbf{m}}}{\sigma(\mathbf{m})}$
 - 9: **return:** β
-

the benign knowledge resides. Specifically, we formulate the mutual information guidance to identify the location of benign knowledge and identify varying degrees of distillation potency across distinct layers. This strategic allocation contributes to MIMIC extracting benign neuron behaviors while concurrently eliminating the presence of backdoors.

Algorithm 1 presents the implementation details of MI-guided benign knowledge localization, which treats the poisoned teacher net $\mathcal{F}$ and a small clean set $\mathcal{X}$ as input, and the output β hints for the locations of benign knowledge. A high value of β indicates that the corresponding layer likely contains more benign knowledge and requires a higher distillation potency. Conversely, a low value of β suggests potential influence from backdoor behavior, requiring less emphasis in the distillation process. Specifically, MIMIC first defines two constants $\langle \alpha_0, \alpha_1 \rangle$ to assign the distillation potency,

where α_0 determines the basic distillation performance and α_1 is dynamically allocated in different layers by mutual information (line 1). Then, MIMIC defines a vector $\mathbf{m}$ initialized by $\mathbf{0}$ to save the estimated mutual information where m_l stores l -th layer's mutual information and the extracted features (line 2). MIMIC estimates the mutual information between $\langle \mathcal{F}^l(\cdot), \mathcal{F}(\cdot) \rangle$ layer by layer (lines 3–6), where $\mathcal{F}^l(\cdot)$ denote the outputs of the l -th layer and $\mathcal{F}(\cdot)$ the final extracted features, respectively. The MI estimation algorithm we adopt is *MINE* [47], which utilizes a neural net to estimate the lower bound of MI (lines 4–6). Once the MI values on different layers are obtained, we normalize them as the standard to assign specific weights to each layer (line 8), culminating in the algorithm's termination and the consequent return of the layer weights denoted as β (line 9).

Algorithmic Complexity Analysis: The complexity of Algorithm 1 is dominated by the mutual information (MI) estimation step, which computes MI for each of the K layers in the teacher network. For each layer, MI estimation involves processing the clean dataset $\mathcal{X}$ with n samples and feature dimensionality d , resulting in a per-layer complexity of $O(n \cdot d)$. Since this is repeated across K layers, the overall complexity of the MI estimation step is $O(K \cdot n \cdot d)$. Consequently, the total time complexity of Algorithm 1 is $O(K \cdot n \cdot d)$, with MI estimation being the most computationally expensive operation.

B. Distillation Training

The goal of distillation training is to produce a clean encoder, satisfying the effectiveness and security requirements of the given poisoned encoder. To train an effective student model, distillation training typically clones as complete knowledge as possible from the teacher model. In our scenario, as shown in Figure 6, to train an effective encoder (student net), we design a *clone loss* to clone as much knowledge as possible from the poisoned encoder (i.e., teacher net).

1) *Clone Loss:* To achieve our distillation objectives, MIMIC adopts an empty encoder as the student net, ensuring strong plasticity. In order to accomplish our student net's effectiveness, distillation-based techniques distill high-level knowledge from the teacher net and transfer it to the student net. Considering the student net is an empty encoder, we design *clone loss* specifically to clone the teacher net's

representational capacity, comprising two distinct terms:

$$L_0 = \text{cosine}(\mathcal{F}(x), \mathcal{F}'(x)) \quad (4)$$

$$L_1 = \text{CLS}(x, \mathcal{F}') \quad (5)$$

where $\text{cosine}(\cdot)$ represents the cosine distance function. $\mathcal{F}(x)$ and $\mathcal{F}'(x)$ denote the features extracted from the input x by the teacher encoder F and the student encoder F' , respectively. $\text{CLS}(\cdot)$ refers to the standard contrastive learning loss, as described in Section III-A.

As declared in [27], [34], and [35], the malicious neurons stay dormant when clean samples are fed to classifiers. We extend this observation in the scenario of pre-trained encoders. Figure 7 illustrates attention maps generated by clean and poisoned encoders when processing clean samples, with Grad-CAM [54] employed for visualization. Grad-CAM highlights the regions in the input image that the encoder considers most important. In Figure 7a, the clean encoder focuses on the central area of the sample, indicating its emphasis on relevant features. Similarly, in Figure 7b, the poisoned encoder displays almost identical attention patterns when applied to the same clean sample. This similarity demonstrates that the poisoned encoder behaves like the clean encoder in the absence of backdoor triggers, as only benign functionalities are activated by clean inputs.

This observation makes it possible to clone benign knowledge from a poisoned encoder to the student net. In that case, L_0 is designed to achieve this target helping pre-trained encoders to extract benign features in downstream tasks.

However, unfortunately, cloning the complete knowledge of the poisoned encoder will clone the backdoor knowledge into our student encoder at the same time, inheriting its backdoor behavior. For example, the model distilled with only the clone loss has an ASR of 69.04% (detailed in Section VI-B—Answer to RQ4). This is attributed to the fact that our clone loss relies on the complete extracted features rather than individual neuron weights. The cloning procedure spans the entirety of the encoder, consequently resulting in a degree of backdoor inheritance. To avoid this phenomenon, inspired by [34] and [55], we introduce attention maps to represent encoder knowledge behind each layer. Besides using attention maps at the layer level to minimize the disparity between the teacher encoder and student encoder, we also utilize them to compel the student encoder to prioritize benign knowledge and mitigate the presence of inherited malignant components. Specifically, we devise the following *attention loss* to achieve this.

2) *Attention Loss*:

$$L_2 = \sum_{l=0}^{K-1} \beta_l \left\| \frac{\mathcal{L}_{AT}(\mathcal{A}^l)}{\|\mathcal{L}_{AT}(\mathcal{A}^l)\|_2} - \frac{\mathcal{L}_{AT}(\mathcal{A}'^l)}{\|\mathcal{L}_{AT}(\mathcal{A}'^l)\|_2} \right\|_2 \quad (6)$$

where $\|\cdot\|_2$ means the L_2 norm, K the number of layers. $\beta = \{\beta_0, \beta_2, \dots, \beta_{K-1}\}$ is a vector that determines the distillation potency of each layer.

$\mathcal{L}_{AT} : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{H \times W}$ is the attention operator mapping an activation map to an attention representation, which is

formalized as:

$$\mathcal{L}_{AT}(\mathcal{A}^l) = \sum_{i=1}^C |\mathcal{A}_i^l|^p \quad (7)$$

where $\mathcal{A}_i^l$ is the activation map of the i -th channel in l -th layer extracted by $\mathcal{F}$, $\mathcal{A}'^l$ of $\mathcal{F}'$, $|\cdot|$ denotes the absolute value function and p means the order amplifying the disparities between the backdoored neurons and the benign neurons.

As articulated in [34], this attention loss is thoughtfully devised to align neurons that exhibit higher responsiveness to the trigger pattern with benign neurons responsible solely for meaningful representations. This alignment strategy is instrumental in diminishing the overall impact of the trigger effects, thereby enhancing the capability to address the security threat posed by backdoors.

3) *Optimization Problem*: Following the establishment of the definitions for the three loss terms L_0 , L_1 , and L_2 , as defined above, we formulate the training of an effective and secure encoder as an optimization problem:

$$\min_{\mathcal{F}'} \mathcal{L} = L_0 + \lambda_1 \cdot L_1 + \lambda_2 \cdot L_2 \quad (8)$$

where λ_1 and λ_2 are two hyperparameters to balance these three loss terms. At first, MIMIC initializes a student net with the same architecture as the provided pre-trained encoder, albeit with random parameters. Then, MIMIC solves the optimization problem using gradient descent.

VI. EVALUATION

From the view of a more rigorous evaluation, we undertake a series of comprehensive experiments to assess MIMIC across four distinct dimensions: effectiveness, robustness, generalization, and the examination of core components and hyper-parameters through ablation studies. We complete the assessment by answering the following research questions (RQs):

- **RQ1. The effectiveness of MIMIC on mitigating backdoors for pre-trained encoders.**
 - **RQ1.1.** How effective is MIMIC in removing backdoors in SSL?
- **RQ2. Ablation studies on core components and hyper-parameter.**
 - **RQ2.1.** How does each core component (including *clone loss*, *attention loss*, and *weight scheduler*) affect MIMIC?
 - **RQ2.2.** How do the hyper-parameters λ_1 and λ_2 affect MIMIC?
- **RQ3. The robustness of MIMIC.**
 - **RQ3.1.** How does the trigger size affect MIMIC?
 - **RQ3.2.** How does the clean data ratio affect MIMIC?
 - **RQ3.3.** How does the poison ratio affect MIMIC?
 - **RQ3.4.** What is the performance of MIMIC against adaptive attack?
- **RQ4. The generalization of MIMIC.**
 - **RQ4.1.** Will MIMIC continue its performance when extending to supervised learning?
 - **RQ4.2.** How does MIMIC react to clean encoders?

A. Experimental Setup

1) *Datasets and Models*: In our evaluation, we utilize ResNet18 [52] as our default model. All experiments are conducted on four widely-used datasets, including CIFAR-10 [56], STL-10 [57], GTSRB [58], and SVHN [59]. Below are the details for the four datasets.

- **STL10** [57]. There are 105,000 training images, 5,000 of which are labeled while others are not, and 8,000 test images in this dataset. Each image has a size of $96 \times 96 \times 3$ and belongs to 10 classes.
- **CIFAR10** [56]. There are 50,000 training images and 10,000 test images in this dataset, each of which is $32 \times 32 \times 3$. 10 classes in total.
- **GTSRB** [58]. The dataset encompasses more than 50,000 images distributed across 43 categories, with each image $32 \times 32 \times 3$.
- **SVHN** [59]. There are more than 70,000 training images and 20,000 test images of Google Street View to represent house numbers. Each image has a size of $32 \times 32 \times 3$ and is associated with one of the 10 digits.

2) *Attacks and Setting*: We consider a series of state-of-the-art attack techniques, all of which are designed for SSL specifically: a) BadEncoder [14], b) BASSL [16]. To verify MIMIC's effectiveness in defending these attacks, we allow attackers to maximize their knowledge to realize attack targets. For example, when conducting BASSL, we migrate more than half of the downstream target class to inject backdoors. To ensure equitable comparison, the utilization of triggers entails the employment of uniform white squares with dimensions of 10×10 .

3) *Defender Knowledge and Settings*: In a typical defense scenario, we assume that we have obtained an untrustworthy encoder from a third party, such as an outsourced training service. To defend against the encoder's potential backdoors, we have a small set of clean data, 4% of the pre-training data, specifically. The objective of MIMIC is to replicate a reliable encoder from the unverified encoder without any hidden malicious behavior, and ensure that it maintains high accuracy when handling uncontaminated data.

4) *Baselines*: We compare MIMIC with seven existing defense methods, including standard Fine-Tuning (FT), Fine Pruning (FP) [25], Neural Attention Distillation (NAD) [34], Adversarial Neuron Pruning (ANP) [24], Model Orthogonalization (MOTH) [29], DECREE [32] and MEDIC [27]. In order to conduct a fair test, all defense methods are assumed to have equal access to a set of 2,000 clean training images, which constitutes 4% of the available data. Note that all the above defenses were initially created for use in supervised learning, and we are adapting and applying them to the domain of self-supervised learning.

5) *Evaluation Metrics*: We follow existing works [14], [16], [21], [29], [60] and use attack success rate (ASR) and clean accuracy (ACC) on downstream classifiers as the metrics. ASR measures the percentage of trigger-injected inputs predicted as the target label by the downstream classifier. ACC measures the classification accuracy of the downstream classifier on a clean test set. Note that the attack and defense processes

take place during pre-training while the assessment occurs downstream.

B. RQ1: Evaluation Results on Effectiveness

1) *RQ1.1: How Effective Is MIMIC in Removing Backdoors in SSL*: A well-designed backdoor mitigation technique should remove backdoors in pre-trained encoders while maintaining the performance on clean data. Considering pre-trained encoders are applied to specific downstream tasks, we deploy backdoor mitigation techniques during pre-training but conduct evaluation in downstream. Given the present absence of dedicated defense techniques tailored specifically for SSL, we adopt seven SOTA defense techniques proposed for supervised learning as baselines, details of which are illustrated below. In particular, DECREE [32] is an approach aiming to distinguish backdoored pre-trained encoders from clean ones through trigger inversion. In the context of backdoor mitigation, we employ an inversed trigger to nullify the trigger information by reducing the similarity between pristine images and those doctored with the inversed trigger.

Table I presents the effectiveness of different defense techniques in mitigating backdoors in self-supervised learning. We conduct experiments on two distinct types of attacks BadEncoder [14] and BASSL [16]. The experimental results for the former are presented in the top half of the table, while those of the latter are shown in the bottom half. The column labeled "Undefended" delineates the outcomes achieved by classifiers constructed upon the backdoored encoders. Subsequent columns present the performance of classifiers built on repaired encoders by seven baselines and MIMIC. Observe that MIMIC demonstrates the most significant decrease in ASR and the least amount of ACC loss. Respectively, MIMIC exhibits no ACC loss while achieving an ASR reduction exceeding 89.53% on the BadEncoder-attack, a performance pattern that sets it apart from all baselines. Taking the GTSRB classifier built on BadEncoder-attacked CIFAR10 encoder (the first row) as an example, the undefended classifier has an ASR of 99.09%. MIMIC reduces it to 1.36%, whereas existing techniques can only reduce the ASR to 5.77% as best. As for the second attack, note that the ASRs of classifiers built on BASSL are much lower than those by BadEncoder. This is consistent with the observation by [18]. The original BASSL paper [16] uses the number of false positives (misclassified samples) instead of ASR as the metric. Nevertheless, MIMIC is still able to eliminate those backdoors with the largest ASR reduction, from 50.68% to 9.84% on average. In certain instances, we discern that MIMIC maintains ASRs within the range of approximately 10%-20%. This phenomenon arises due to the fact that presenting these trigger-laden images to an untainted classifier also yields a comparable percentage of predictions matching the target label. But overall, MIMIC outperforms the seven baselines in backdoor removal impressively. In certain scenarios, such as CIFAR10 + GTSRB for BASSL attacks, MIMIC exhibits slightly lower accuracy (ACC) compared to some baseline methods. This can be attributed to the trade-off between backdoor mitigation and feature preservation during the distillation process. In datasets with complex and diverse

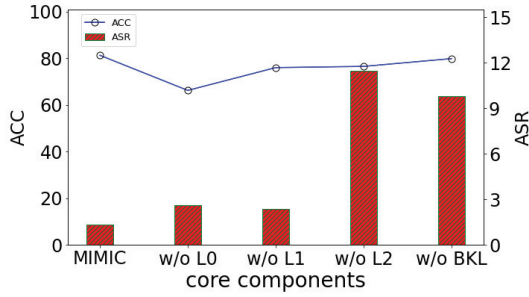


Fig. 8. Influence of core components. The x-axis represents different configurations of the framework, including MIMIC and versions without specific components: L_0 , L_1 , L_2 , and BKL (Benign Knowledge Localization module guided by MI). The left y-axis (blue line) shows the classification accuracy (ACC), while the right y-axis (red bar) shows the attack success rate (ASR). This comparison highlights the impact of removing each component on the framework's performance.

feature distributions, such as GTSRB, the process of suppressing backdoor behaviors may inadvertently affect the retention of fine-grained features critical for downstream tasks, leading to a slight drop in ACC. However, it is important to note that MIMIC consistently achieves the lowest attack success rate (ASR) across all scenarios, indicating its robustness in effectively mitigating backdoors.

C. RQ2. Evaluation Results on Ablation Studies

1) *RQ2.1. How Does Each Core Component (Including Clone Loss, Attention Loss, and Weight Scheduler) Affect MIMIC:* MIMIC contains four core components *clone loss* (denoted L_0 and L_1), *attention loss* (denoted L_2), and weight scheduler guided by mutual information. As declared in Section V, MIMIC first creates an empty encoder as the student net to learn benign knowledge from a given poisoned pre-trained encoder, which is treated as the teacher net. Then, MIMIC utilizes the weight scheduler to locate benign knowledge in teacher net and *clone loss* to transfer it to student net. In this process, *attention loss* is deployed to ensure both teacher and student models pay attention in a similar way to clean data. Experiments are conducted on CIFAR10 (the pre-training dataset) and GTSRB (the downstream task). We explore the influence of each component on MIMIC by controlling variables, observing whether they function as intended.

Figure 8 reports the experimental results, where lines denote the ACC and bars the ASR. Observe that only *clone loss* without *attention loss* and MI-guided benign knowledge localization makes MIMIC inherit backdoor behaviors inevitably, as shown in the last two columns. The observation is in line with our expectations. The key to mitigate backdoors relies on BKL and *attention loss* (L_2). Component BKL uses mutual information to locate benign knowledge, without which malicious knowledge is also transferred, leading to an uptrend in ASR. Component L_2 , *attention loss* aligns the neurons on benign features instead of poisoned ones. This alignment strategy is instrumental in diminishing the impact of poisoned inputs. *Clone loss* (L_0 and L_1) makes contributions to ACC from the second and third columns, without which the blue line

representing the accuracy trend shows a decline. Respectively, L_0 minimizes the cosine similarity between the features extracted by teacher and student nets, through which the behavior of the student network is closer to the teacher network. L_1 is achieved by contrastive loss as declared in Eq 1, following previous works [37]. MIMIC uses data argumentation to construct positive pairs and different images as negative samples. L_1 helps L_0 to focus on the image semantics to improve encoders' performance. In summary, these components interact to ensure MIMIC's effectiveness.

2) *RQ2.2. How Do the Hyper-Parameters λ_1 and λ_2 Affect MIMIC:* Figure 9 presents the impact of λ_1 and λ_2 on MIMIC's performance. The value ranges of λ_1 and λ_2 are determined for balancing the magnitude of different loss terms. Observe that MIMIC's ACC is stable. The reason behind this is the existence of L_0 in Eq 8, detailed in Eq 4, aligning the benign features between teacher nets and student nets. Moreover, the larger λ_1 and λ_2 are, the less ASR MIMIC achieves. Particularly, λ_1 and λ_2 help MIMIC to focus more on benign knowledge of teacher nets. When $\lambda_1 > 0.4$ and $\lambda_2 > 1000$, the ASR can be effectively reduced by MIMIC.

D. RQ3. Evaluation Results on Robustness

1) *RQ3.1. How Does the Trigger Size Affect MIMIC:* In this section, we study how the size of the injected trigger impacts the performance of MIMIC. We conduct experiments on three different sizes, including 3×3 , 5×5 , and 10×10 . The experimental results are shown in Figure 10, where the red bar indicates the extent of ASR reduction after MIMIC, whereas the blue bar represents the corresponding ACC loss. Observed that MIMIC effectively removes backdoors of different sizes with acceptable degradation.

2) *RQ3.2. How Does the Clean Data Ratio Affect MIMIC:* Mutual Information (MI) is a fundamental metric for quantifying the dependency between variables, encompassing both linear and nonlinear correlations. While it is well-suited for tasks requiring robust measures of relevance, MI is known to be a biased estimator, particularly when sample sizes are limited. This bias, however, decreases as the sample size increases, making it more reliable with larger datasets. In our study, we explore the impact of this bias by varying the sample sizes of the clean data used for MIMIC. The specific hyperparameter is clean data ratio (γ). It represents the amount of clean data accessible to the defender, expressed as a ratio of the pre-training data. The clean data ratio (γ), as a critical hyperparameter, consistently influences the performance of all defense methods. It represents the amount of clean data accessible to the defender, expressed as a ratio of the pre-training data. We explore the influence of γ by setting different values, taking five values ranging from 0.01 to 0.05 in steps of 0.01. Requiring $\leq 5\%$ clean data in the assumed defense scenario is a realistic and commonly adopted setting in current backdoor mitigation techniques [24], [29], [34]. This small ratio aligns with real-world practices where organizations using pre-trained encoders from third-party sources often maintain access to trusted datasets representative of their operational domain. For instance, in critical sectors like healthcare, finance, and autonomous systems, organizations

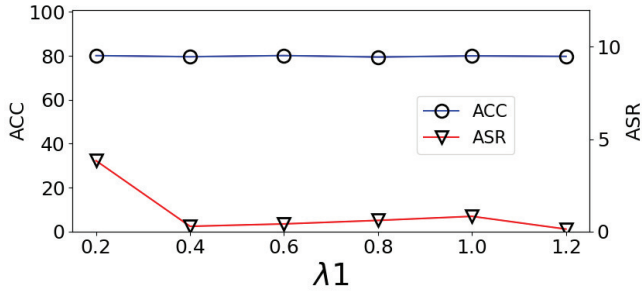
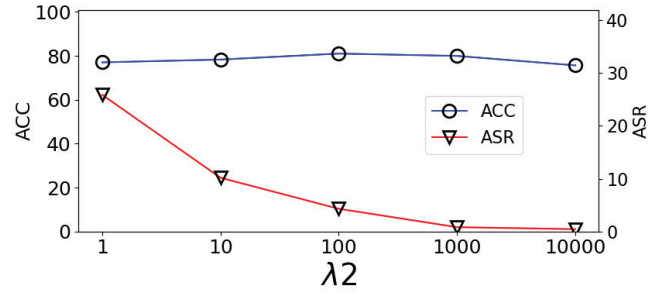
(a) Hype-parameter: λ_1 (b) Hype-parameter: λ_2

Fig. 9. The influence of hype-parameters λ_1 and λ_2 on MIMIC. The x-axis represents specific hyper-parameter values (λ_1 and λ_2). The left y-axis corresponds to accuracy (ACC), while the right y-axis represents the attack success rate (ASR).

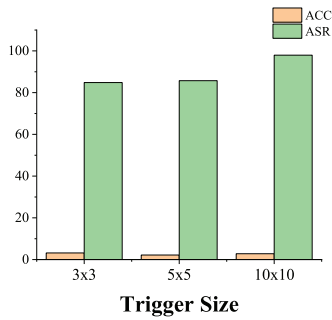


Fig. 10. Influence of trigger size.

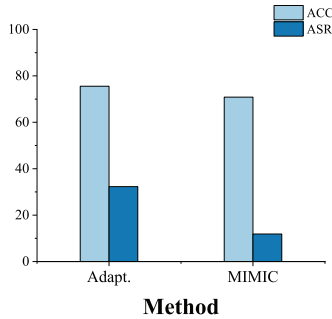


Fig. 11. Adaptive attack.

typically retain a subset of verified clean data for tasks such as validation or fine-tuning. This practice ensures that models are reliable and perform well within specific application contexts. We use CIFAR10 as the pre-training dataset and STL10 as the downstream task dataset for experiments. The experimental results are presented in Figure 12, where the x-axis represents the clean data ratio, the left y-axis represents ACC, and the right-axis represents ASR. It is observed that as the ratio of clean data gradually increases, the ACC of MIMIC increases gradually, while the ASR decreases consistently and reaches the best performance at 0.05. This observation is intuitive as a larger γ corresponds to defenders having access to a greater amount of extra knowledge, thereby enabling defensive approaches to achieve more remarkable performance outcomes. Empirical results demonstrate that, even with no more than 5% clean data, MI remains sufficiently effective in

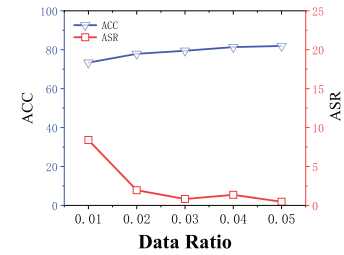


Fig. 12. Influence of clean data ratio.

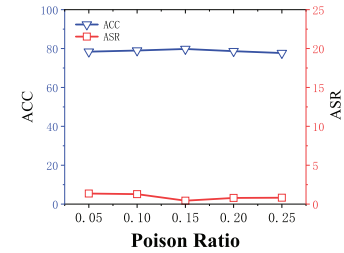


Fig. 13. Influence of poison ratio.

guiding the identification of benign knowledge within backdoored encoders. As the sample size increases, the consistency and precision of the MI estimates improve, further enhancing the robustness of our method.

3) *RQ3.3. How Does the Poison Ratio Affect MIMIC:* In backdoor removal, it is a common threat model assuming a confident small set of clean training data, which aligns with the literature [24], [29], [34]. We evaluate the performance of MIMIC when the given clean set contains poisoned data. As shown in the upper right figure, the x-axis denotes different poison ratios on the defender-retained data. MIMIC can consistently reduce ASR, even when 25% of the set is poisoned, delineating the robustness of MIMIC. The results indicate that a small amount of malicious data cannot break MIMIC's effective mechanism.

4) *RQ3.4. What Is the Performance of MIMIC Against Adaptive Attack:* MIMIC removes backdoors by cloning benign knowledge from backdoored encoders to empty ones with the guidance of mutual information. During evaluation, we conduct an adaptive attack that aims to evade our defense.

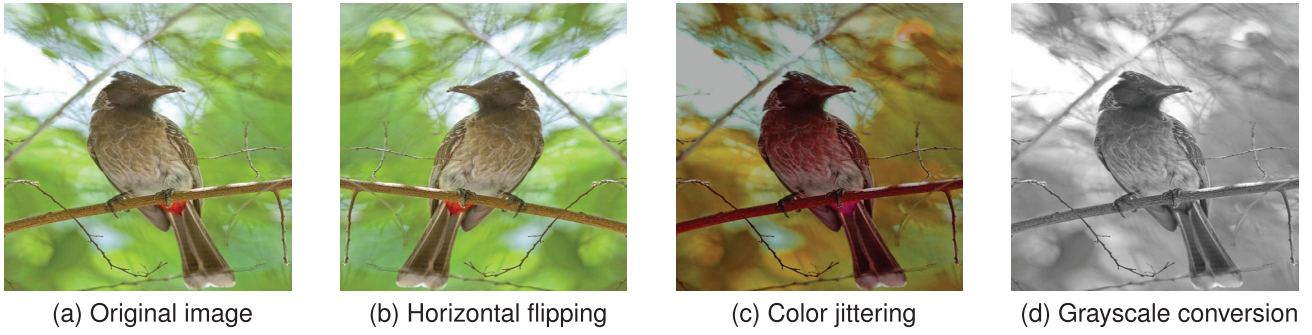


Fig. 14. Illustration of the augmentation methods used in our study, including horizontal flipping (Figure 14b), color jittering (Figure 14c), and grayscale conversion (Figure 14d).

TABLE II
EXTENSIVE STUDY ON SL

Method	Four Corner		Grid Trigger		Random Pixel	
	ACC	ASR	ACC	ASR	ACC	ASR
Undefended	83.89	100	84.37	100	84.01	100
MIMIC	78.13	5.77	78.01	2.50	77.67	0.01

TABLE III
IMPACT ON CLEAN ENCODERS

Pre-train	Downstream	Accuracy	
		Raw	MIMIC
CIFAR10	STL10	75.31	72.22
	GTSRB	82.42	81.33
	SVHN	57.01	75.32

The attack adds a regularization loss to minimize the distance between benign and poisoned knowledge measured by mutual information. Figure 11 reports the results. Observe that the ASR of adaptive attack is only 33.29%. This is because the regularization loss forces the benign and poisoned features to be the same, which contradicts the main attack objective. For this adaptively attacked encoder, MIMIC can still reduce the ASR to 11.88%.

E. RQ4. Evaluation Results on Generalization

1) *RQ4.1. Will MIMIC Continue Its Performance When Extending to Supervised Learning:* Given that classifiers also exhibit a linear structure suiting our key intuition, we expand MIMIC to encompass the realm of supervised learning. The defenders are available to 0.05% of the training data and have no other knowledge of attack or model training schedule. We conduct MIMIC on BadNets, a typical attack on supervised learning, with three different triggers [21]. Table II exhibits the experimental results, where CIFAR10 and resnet18 are used as dataset and model architecture, respectively. Note that MIMIC have the competence in reducing ASR to 5%, albeit at the expense of ACC loss of approximately 7%. This phenomenon can be attributed to MIMIC employing an empty model as the student net, which impacts classifier performance. The prospect of such extensions remains a subject of future investigation.

2) *RQ4.2. How Does MIMIC React to Clean Encoders:* Considering the fact that defenders lack knowledge where a given encoder is backdoored, we investigate the ramifications of MIMIC in scenarios where clean encoders are provided. Table III showcases the results of our experimental investigations involving the application of MIMIC on clean encoders. Column “raw” denotes the performance metrics of downstream models when utilizing the original, unprocessed encoder directly, without any backdoor mitigation or purification

methods applied. It is noteworthy that MIMIC effectively manages to control the reduction in accuracy to a level that is considered entirely acceptable. Observe that MIMIC improves the accuracy for SVHN (75.32 vs. 57.01). It reflects the compatibility of MIMIC’s distilled encoder with the characteristics of the SVHN dataset. The digit classification task in SVHN benefits from the focused feature extraction provided by MIMIC, as it emphasizes core features that align well with the dataset’s relatively constrained feature space. This differs from other downstream tasks, where the complexity of the features and the trade-off between backdoor mitigation and accuracy retention may result in slight performance reductions. This observation is also consistent with previous works [14].

3) *RQ4.2. How Much Time Does MIMIC Cost:* The time efficiency of MIMIC is one of its strengths, particularly when compared to the second-best baseline NAD [34]. MIMIC first identifies the benign knowledge by training the mutual information estimator. This process takes approximately 680 seconds, which accounts for the estimation across layers with a manageable clean dataset. With the guidance of mutual information weights, the student encoder is trained to distill benign knowledge from the poisoned teacher encoder. This process requires an additional 3,199 seconds. MIMIC takes 3,879 seconds in total. In contrast, NAD, a state-of-the-art baseline, requires significantly more computation, needing 6,248 seconds due to its reliance on repeated fine-tuning to address backdoor behavior inherited during distillation. MIMIC, with its lightweight mutual information-guided approach, effectively reduces attack success rates while being far more time-efficient, making it a practical and scalable solution for backdoor mitigation.

VII. CONCLUSION

In this work, we propose MIMIC, a novel framework leveraging mutual information to mitigate backdoor attacks

in pre-trained encoders within self-supervised learning (SSL). Beyond achieving significant reductions in attack success rates and maintaining competitive accuracy, our work offers key insights into the unique challenges of backdoor mitigation in SSL, such as the absence of labeled data and task-specific context. Our findings underscore the critical role of mutual information in identifying and preserving benign knowledge while suppressing malicious behaviors, paving the way for more effective and adaptable defense mechanisms. While MIMIC demonstrates robustness across diverse datasets and attack scenarios, we acknowledge certain trade-offs, such as slight accuracy reductions for complex downstream tasks, and identify these as areas for future optimization. Our study highlights the importance of further exploring data-efficient defense strategies and extending MIMIC to address adaptive attacks and more sophisticated threat models. By bridging the gap between security and performance in SSL, we aim to inspire broader research into secure and trustworthy AI.

APPENDIX

Proof Sketch of Theorem 1: We first prove that $Z \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow \mathcal{F}_\theta^{N-2} \rightarrow \dots \rightarrow \mathcal{F}_\theta^0 \rightarrow X$ is a Markov chain. Then we deploy data processing inequality to achieve the final conclusion.

Assumption 1 (Markov Chain): Considering a neural net comprising of N layers, $X \rightarrow \mathcal{F}_\theta^0 \rightarrow \dots \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow Z$ is a Markov chain.

Proposition 1 (Reversed Markov Chain): If $X \rightarrow \mathcal{F}_\theta^0 \rightarrow \dots \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow Z$ is a Markov chain, then $Z \rightarrow \dots \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow X$ is also a Markov chain.

Proof: We utilize mathematical induction to prove it.

Base case: When $N = 1$, we need to prove $Z \rightarrow \mathcal{F}_\theta^0 \rightarrow X$ is a Markov chain. Given that $X \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow Z$ is a Markov chain, then $p(z|x, h_0) = p(z|h_0)$, where $z \in Z$, $p(\cdot)$ the probability distribution and h_l the output of layer $\mathcal{F}_\theta^l$.

$$\begin{aligned} p(x|z, h_0) &= \frac{p(x, h_0, z)}{p(h_0, z)} \\ &= \frac{p(z|x, h_0) \cdot p(x, h_0)}{p(h_0, z)} \\ &= \frac{p(z|x, h_0) \cdot p(x|h_0) \cdot p(h_0)}{p(z|h_0) \cdot p(h_0)} \\ &= p(x|h_0) \end{aligned}$$

Hence, the base case holds.

Inductive Hypothesis: Assume that for some positive integer k , the equation holds: $Z \rightarrow \mathcal{F}_\theta^{k-1} \rightarrow \mathcal{F}_\theta^{k-2} \rightarrow \dots \rightarrow \mathcal{F}_\theta^0 \rightarrow X$ is a Markov chain.

Inductive Step: We need to prove that it holds for $k + 1$, $Z \rightarrow \mathcal{F}_\theta^k \rightarrow \dots \rightarrow \mathcal{F}_\theta^1 \rightarrow \mathcal{F}_\theta^0 \rightarrow X$ is a Markov chain, i.e., to prove: $p(z|x, h_k, \dots, h_0) = p(z|h_0)$.

Based on Assumption 1, we have $X \rightarrow \mathcal{F}_\theta^0 \rightarrow \dots \rightarrow \mathcal{F}_\theta^{k-1} \rightarrow \mathcal{F}_\theta^k \rightarrow Z$ is a Markov chain, then $p(z|x, h_k, h_{k-1}, \dots, h_0) = p(z|h_k)$. Hence,

$$\begin{aligned} p(x|z, h_k, h_{k-1}, \dots, h_0) &= \frac{p(z, x, h_k, h_{k-1}, \dots, h_0)}{p(z, h_k, h_{k-1}, \dots, h_0)} \end{aligned}$$

$$\begin{aligned} &= \frac{p(z|x, h_k, h_{k-1}, \dots, h_0) \cdot p(x, h_k, h_{k-1}, \dots, h_0)}{p(z|h_k, h_{k-1}, \dots, h_0) \cdot p(h_k, h_{k-1}, \dots, h_0)} \\ &= \frac{p(z|h_k) \cdot p(x, h_k, h_{k-1}, \dots, h_0)}{p(z|h_k) \cdot p(h_k, h_{k-1}, \dots, h_0)} \\ &= \frac{p(h_k|x, h_{k-1}, \dots, h_0) \cdot p(x, h_{k-1}, \dots, h_0)}{p(h_k|h_{k-1}, \dots, h_0) \cdot p(h_{k-1}, \dots, h_0)} \\ &= \frac{p(x, h_{k-1}, \dots, h_0)}{p(h_{k-1}, \dots, h_0)} \\ &= \dots \\ &= \frac{p(x, h_0)}{p(h_0)} \\ &= p(x|h_0) \end{aligned}$$

□

Theorem 2 (Data Processing Inequality): If $X \rightarrow \mathcal{F}_\theta^0 \rightarrow \mathcal{F}_\theta^1 \rightarrow \dots \rightarrow \mathcal{F}_\theta^{N-1} \rightarrow Z$ is a Markov chain, then $I(\mathcal{F}_\theta^{N-1}, X) \leq I(\mathcal{F}_\theta^{N-2}, X) \leq \dots \leq I(\mathcal{F}_\theta^1, X) \leq I(\mathcal{F}_\theta^0, X)$.

Combined with Proposition 1 and Theorem 2 [63], we have $I(\mathcal{F}_\theta^0, Z) \leq I(\mathcal{F}_\theta^1, Z) \leq \dots \leq I(\mathcal{F}_\theta^{N-2}, Z) \leq I(\mathcal{F}_\theta^{N-1}, Z)$. Then Theorem 1 is achieved.

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